

Assignment 2

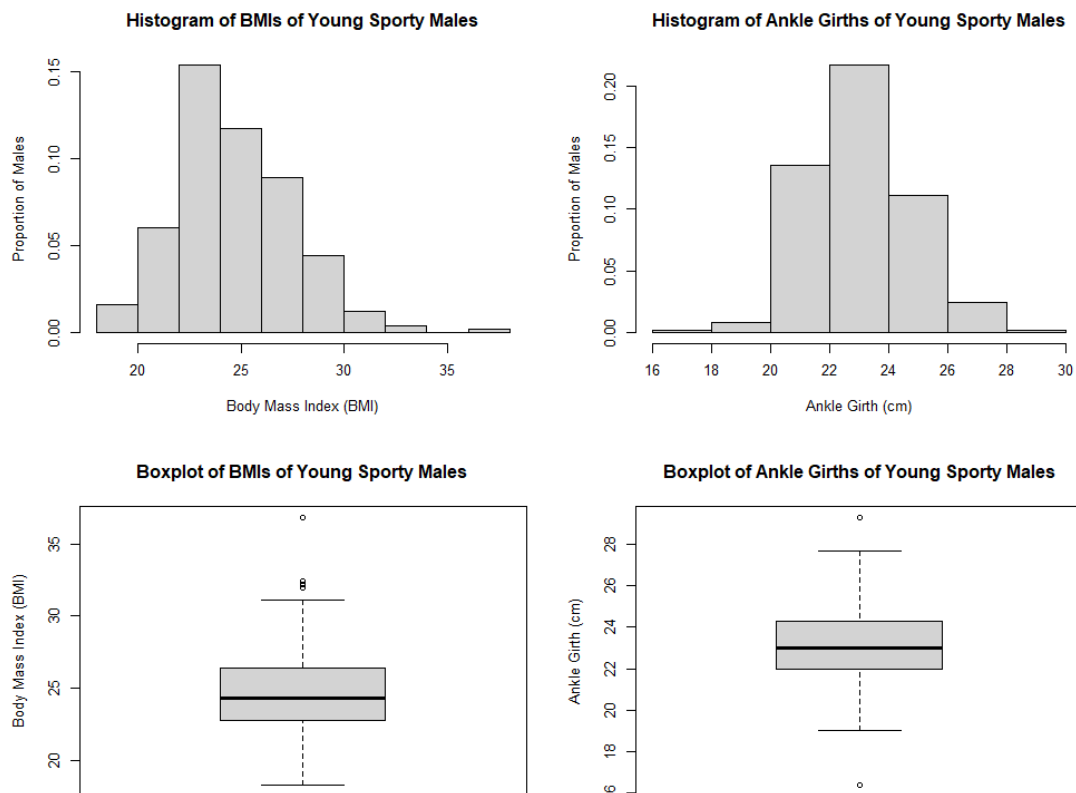
Exercise 2.4

Part a.

Consider the plots below. The histogram of BMIs suggests that the distribution of BMI is asymmetric with a long right tail and a fat left tail. The histogram of ankle girths below suggests that the distribution of ankle girths is bell-shaped. Higher proportion of values (0.14, 0.22, 0.11) around the mean in the histogram of ankle girths compared to smaller proportion of values (0.15, 0.12, 0.8) around the mean in the histogram of BMIs suggests higher spread of the distribution of BMI compared to the distribution of ankle girths.

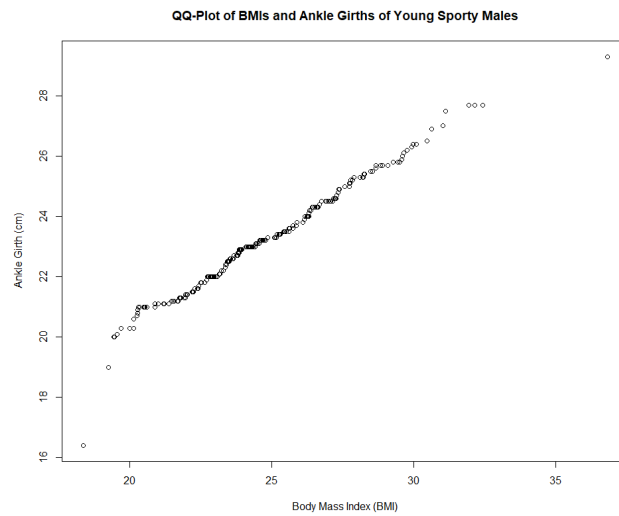
The almost identical boxplots suggest approximately equal shapes of distributions of both BMI and ankle girths. The distributions of BMI and ankle girth seem to be both a little right-skewed as the middle line in boxplots suggests. The whiskers of both boxplots are of similar proportionality to the box length which suggests similar variability of BMI and ankle girth.

As the circles representing extreme values in the boxplots suggest, it seems that the differences in symmetry and spread are caused by a few big extreme BMI values and the absence of small extreme BMI values which make the long tail and higher spread in the BMI data. Therefore, I would dare to conclude that the differences in shape are caused by extreme BMI values, and overall, the distributions of BMI and ankle girth have approximately the same shape.



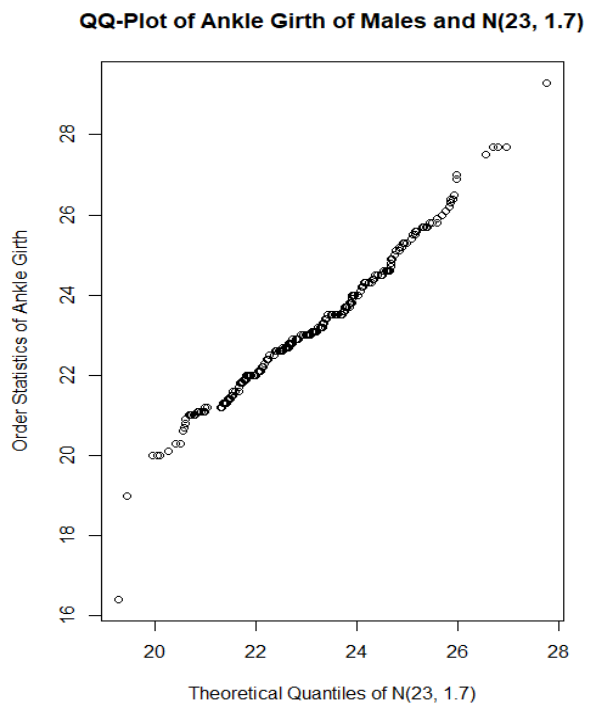
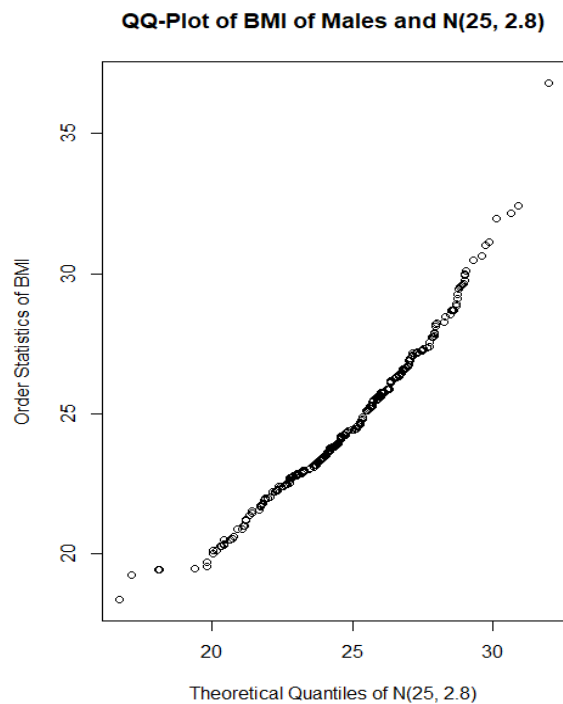
Part b.

The points on a straight line in the QQ-plot below indicate that the underlying distributions of BMI and ankle girth belong to the same location-scale family.



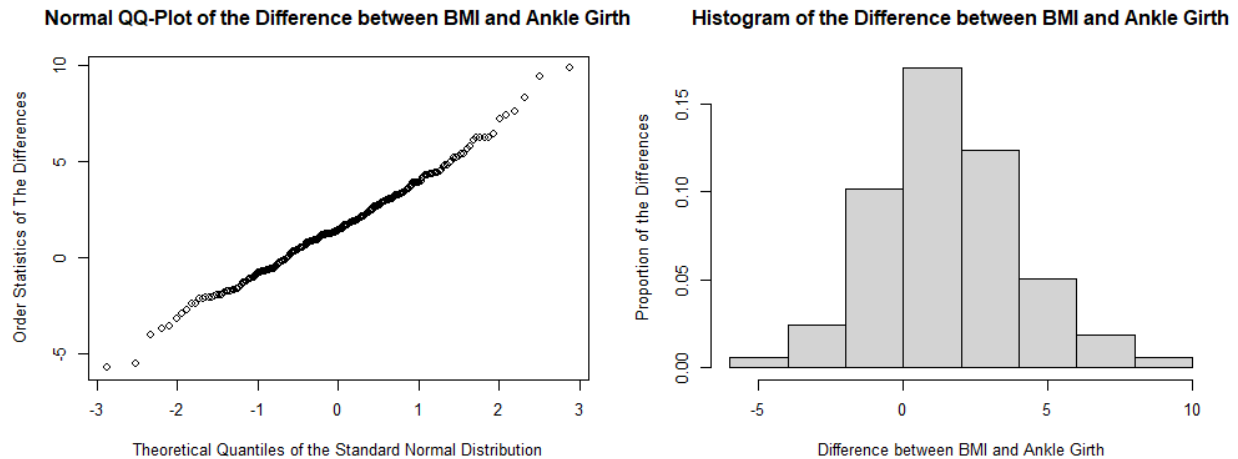
Part c.

We examined the location-scale family of normal distributions. The QQ-plots below show points on the line $y = x$. This suggests that $\text{BMI} \sim N(25, 2.8)$ and $\text{ankle girth} \sim N(23, 1.7)$.



Part d.

Judging by the straight line in the QQ-plot below and the bell-shape of the histogram below, the normality may be assumed for the differences between BMI and ankle girth.



Part e.

The Shapiro-Wilk normality test gives $W = 0.99$ and $p\text{-value} = 0.24$. As $p > 0.05$, the null hypothesis, which states that the difference between BMI and ankle girth is normally distributed, cannot be rejected with 95% confidence. This indicates that H_0 may be correct. The high value 0.99 of W very close to 1 indicates that the difference between BMI and ankle girth is normally distributed. These conclusions support the assumption of normality of the difference between BMI and ankle girth.

Part f.

The Shapiro-Wilk test gives $W = 0.98$ and $p\text{-value} = 0.001$, and $W = 0.96$ and $p\text{-value} = 0.095$ for BMI of 247 and BMI of 50 young males, respectively. The test rejects the null hypothesis of normality for the BMI sample of 247 males as the $p\text{-value} < 0.05$, but not for the BMI sample of 50 males as $p\text{-value} > 0.05$. Hence, the greater sample size provided stronger confidence for rejecting normality of BMI. Both histograms below seem to be asymmetric and are not of nice bell-shape typical for normality. Hence, we could conclude that BMI is not normally distributed. The denominator of W contains an estimator for $(n - 1) \sigma^2$. Hence, the greater the sample size, the lower W . The low value of W allows us to reject H_0 . Therefore, the greater sample size provides greater confidence for rejecting normality.

